

KISANKAVACH

Karnataka

System Architecture

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Farmer Credit. Benefits. Protection.

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Note: Page references are represented by section numbering to preserve stability across approved revisions.

1. Purpose and Scope

Defines the Karnataka pilot baseline for farmer-centred Kisan Credit Card enablement, benefit visibility, consent-driven data exchange, grievance handling, fraud protection and programme oversight.

In scope: farmer mobile experience, assisted service, participating-bank workflow, Government command centre, authorised integrations and operational reporting.

2. Guiding Principles

Kannada-first and low-literacy friendly; reuse verified information; explicit purpose-bound consent; least privilege; interoperability through adapters; configurable workflows and auditability.

The bank remains the final credit decision-maker. Government-originated and bank-originated data retain their source and provenance.

3. Architecture Overview



4. Deployment and Network

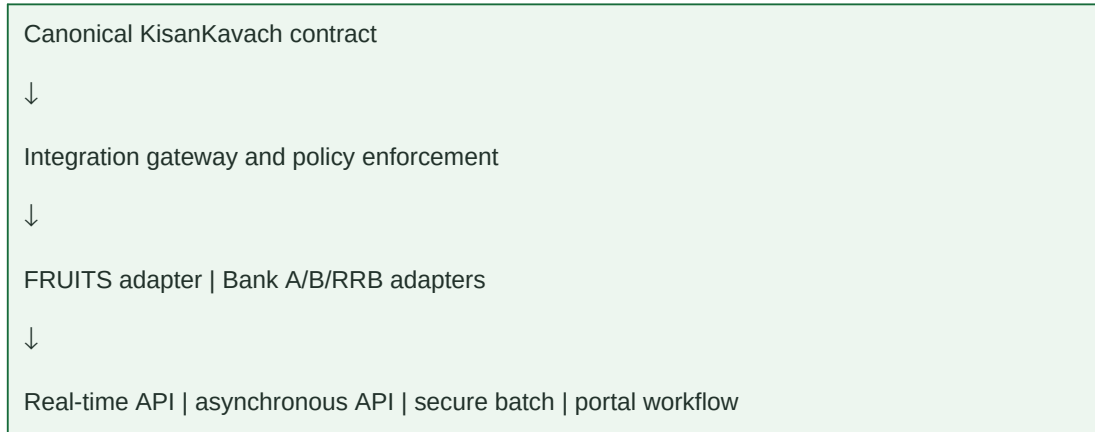
AWS reference deployment uses Route 53, CloudFront, WAF, API Gateway or ALB, private application subnets, isolated data subnets, managed PostgreSQL, Redis, encrypted object storage and protected logs. Development, QA, UAT, pre-production and production are separated.

5. Service Boundaries

Service	Responsibility
Identity	Authentication, tokens, MFA, roles and sessions
Farmer	Profile aggregation, external references and provenance
KCC	Drafts, eligibility guidance, submission, status and renewal
Workflow/SLA	Configurable transitions, timers, alerts and escalation
Document	Metadata, upload, quarantine, scan and retrieval
Consent	Purpose, data categories, recipient, validity and withdrawal

Service	Responsibility
Integration	Canonical contracts, adapters, retry and reconciliation
Audit/Reporting	Evidence, operational metrics and analytical feeds

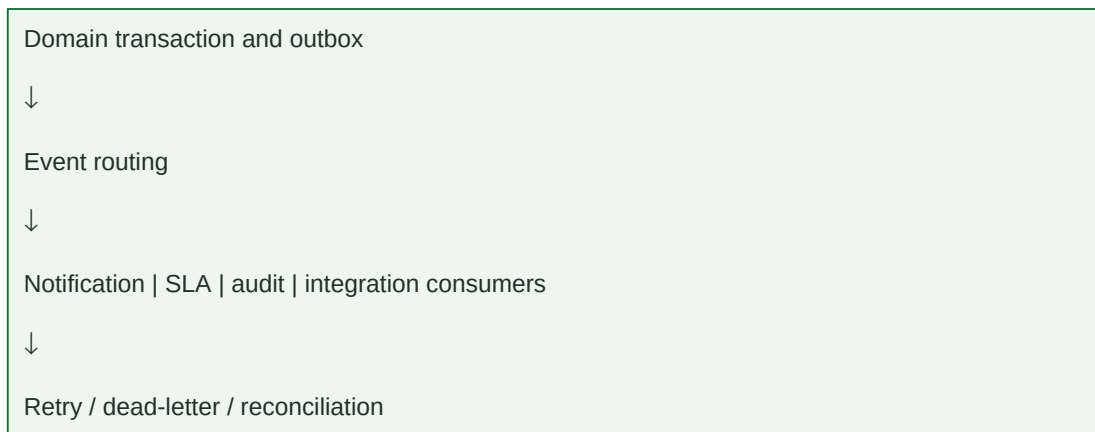
6. Integration Architecture



Adapters isolate core services from partner schema and protocol change. Correlation IDs, idempotency keys, retries with backoff, dead-letter handling and reconciliation are standard.

7. Event and Workflow Architecture

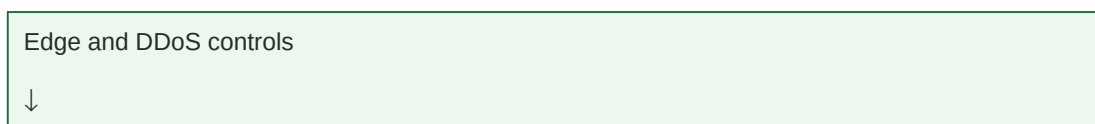
Key events include KCC_APPLICATION_SUBMITTED, DOCUMENT_UPLOADED, BANK_STATUS_CHANGED, KCC_SANCTIONED, KCC_REJECTED, KCC_DISBURSED, SLA_WARNING, SLA_BREACHED, GRIEVANCE_CREATED, FRAUD_DETECTED and CONSENT_WITHDRAWN.

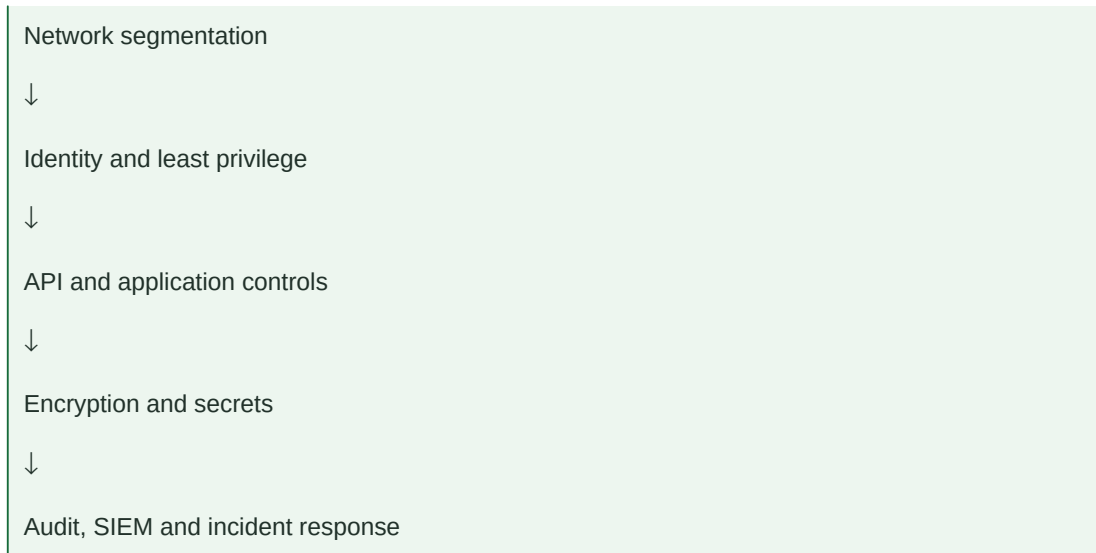


8. Data and Analytics

Operational PostgreSQL is domain-partitioned; Redis is restricted to safe short-lived cache and controls; documents remain in encrypted object storage. An event/ETL pipeline separates Government analytics from transactional workloads.

9. Security Architecture





No public data buckets or Internet-exposed database. Privileged operations require MFA and audit; data exports are scoped, approved and monitored.

10. Availability and Recovery

Concern	Design response
Availability	Multi-AZ managed services and stateless scaling
Failure isolation	Queues, circuit breakers and adapter boundaries
Recovery	Encrypted backups, restore tests, defined RPO/RTO
Deployments	Automated checks, canary/blue-green option and rollback
Observability	SLIs, business metrics, logs, traces and on-call alerts

11. Architecture Decisions

Pilot implementation may use a modular service deployment rather than excessive microservice fragmentation. Canonical APIs and domain boundaries are preserved so high-change or high-scale functions can later be separated.

End of controlled baseline